

St George Girls' High School

Trial Higher School Certificate Examination

2003



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- All questions may be attempted.
- Begin each question on a new page
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (15 marks) – Start a new page

Marks

a) Show that $y = xe^{-x}$ has a stationary point at the point $\left(1, \frac{1}{e}\right)$. 2

b) On separate diagrams sketch the following curves. Indicate clearly any turning points, asymptotes and intercepts with the coordinate axes. 9

(i) $y = xe^{-x}$

(ii) $y = x^2e^{-2x}$

(iii) $y = \frac{1}{x^2e^{-2x}}$

(iv) $y = \log_e(xe^{-x})$

(v) $y = e^{xe^{-x}}$

c) Solve for x:

$$\frac{3-x}{\sqrt{x-1}} \geq \frac{\sqrt{x-1}}{3-x}$$

4

Question 2 – (15 marks) – Start a new page

Marks

a) Evaluate the following definite integrals:

(i) $\int_0^1 \frac{2x}{1+2x} dx$ 3

(ii) $\int_2^3 \frac{x+1}{\sqrt{x^2+2x+5}} dx$ 3

(iii) $\int_2^4 \frac{dx}{x\sqrt{x-1}}$ 3

b) The cubic equation $x^3 - 2x + 4 = 0$ has roots α, β, γ .

(i) Prove that the cubic equation with roots $\alpha^2, \beta^2, \gamma^2$ is $x^3 - 4x^2 + 4x - 16 = 0$. 3

(ii) Factorise $x^3 - 4x^2 + 4x - 16$ into linear factors and hence prove that only one of α, β, γ is real. 3

Question 3 – (15 marks) – Start a new page

Marks

a) Let $P(z) = z^7 - 1$.

(i) Find all the complex roots of $P(z) = 0$. (Call them $z_0, z_1, \dots, z_6$ leaving all answers in terms of π).

3

(ii) Plot the points representing $z_0, z_1, \dots, z_6$ on the Argand Diagram.

(iii) Factorise $P(z)$ over the complex numbers.

1

(iv) Factorise $P(z)$ over the real numbers (leave answers in terms of π).

2

(v) Show that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$

2

b) The area bounded by the curve $y = x(2 - x)$ and the x axis is rotated about the y axis.

5

(i) By considering cylindrical shells with generators parallel to the y axis, show that the volume V units³ of the solid so generated is given by $V = \int_0^2 2\pi xy \, dx$.

(ii) Hence, determine the volume of this solid.

Question 4 – (15 marks) – Start a new page

Marks

a) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 2

(ii) With reference to an appropriate diagram, show geometrically why the above result is true. 2

(iii) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ 3

b) If $I_n = \int_0^1 x^n e^{-x} dx$ ($n \geq 0$)

(i) Prove that $I_n = n I_{n-1} - \frac{1}{e}$ ($n \geq 1$) 2

(ii) Hence, evaluate $\int_0^1 x^3 e^{-x} dx$ 2

c) Using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, calculate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \theta \tan \frac{\theta}{2} d\theta$ 4

Question 5 – (15 marks) – Start a new page

Marks

a) Find all pairs of real numbers x and y that satisfy $(x + iy)^2 = -12 + 16i$

3

b) Consider the polynomial equation

$$z^3 + (6 + i)z^2 + (17 + 8i)z + 33i + 30 = 0$$

(i) This equation has a root $\alpha = pi$ where p is real. Find the value of p .

3

(ii) If the other roots are β and γ , use the relationships for the sum and product of the roots (or otherwise) to find β and γ .

[Hint: Use your answer from part (a)].

3

(iii) Show that the points representing α, β, γ on the Argand diagram are the vertices of a right-angled triangle.

1

c) The complex number z and its conjugate $\bar{z}$ satisfy

$$(z - \bar{z})^2 + 8(z + \bar{z}) = 16$$

5

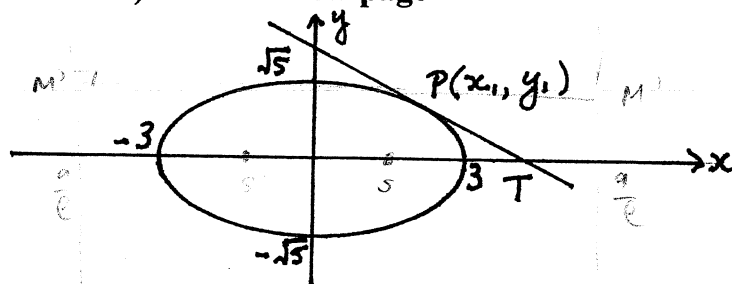
(i) Prove that the point which represents z on the Argand diagram lies on a parabola.

(ii) Sketch the locus of z and hence show that $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$

Question 6 – (15 marks) – Start a new page

Marks

a)



The point $P(x_1, y_1)$ lies on the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ ($x_1 \neq 0$)

The tangent to the ellipse at P cuts the x axis at T .

(i) Find the eccentricity, e , of the ellipse.

1

(ii) Find the coordinates of the foci, S and S'

1

(iii) Find the equations of the directrices.

1

(iv) Show that the equation of the tangent at P is $\frac{x x_1}{9} + \frac{y y_1}{5} = 1$

2

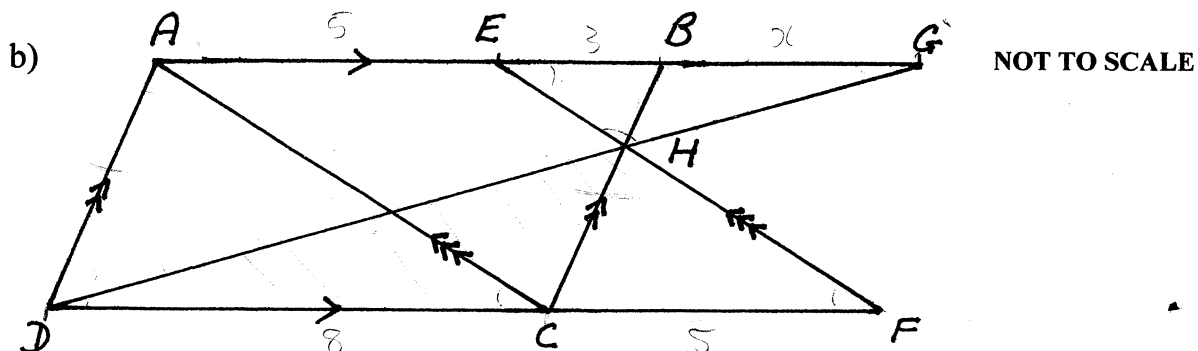
(v) Hence write down the coordinates of T .

1

(vi) Using the focus-directrix definition of the ellipse, or otherwise, show that

4

$$\frac{PS}{PS'} = \frac{TS}{TS'}$$



5

$ABCD$ and $AEFC$ are parallelograms, $AE = 5\text{cm}$ and $EB = 3\text{cm}$.

(i) Find, giving brief reasons, the length of AG .

(ii) Find the ratio of the area of $\triangle CDH$ to the area of the parallelogram $ABCD$.

Question 7 – (15 marks) – Start a new page

Marks

- a) The acceleration of a particle which is moving along the x axis is given by

$$\frac{d^2x}{dt^2} = 2x^3 - 10x$$

- (i) If the particle starts at the origin with velocity u show that its velocity v is given by $v^2 - u^2 = x^4 - 10x^2$

2

- (ii) (α) If $u = 3$ show that the particle oscillates within the interval $-1 \leq x \leq 1$.

4

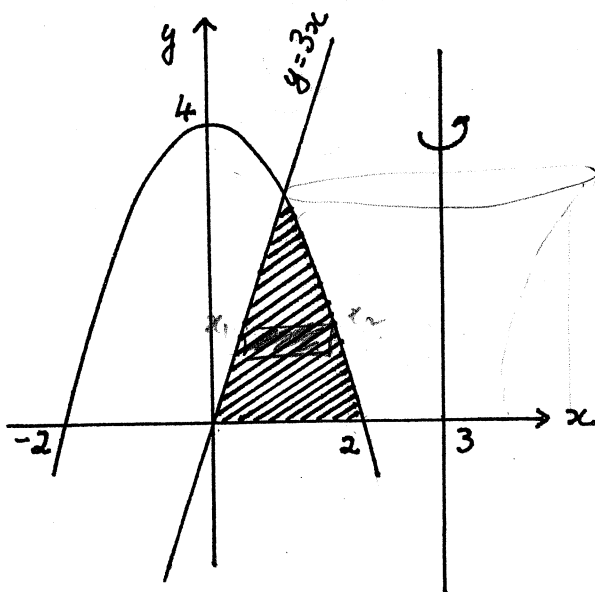
- (β) Is this an example of simple harmonic motion? Give a clear reason for your answer.

1

- (iii) If $u = 6$ carefully describe the motion of the particle.

3

b)



The area enclosed by the curve $y = 4 - x^2$, the line $y = 3x$ and the x axis (for $x \geq 0$) is rotated about the line $x = 3$.

Calculate the volume of the solid generated.

5

Question 8 – (15 marks) – Start a new page

Marks

- a) (i) x, y and z are positive integers. Show that if x is a factor of both y and z then x is a factor of $z - y$.

1

(ii) Show that $2^{2^{k+1}} = \left(2^{2^k}\right)^2$

1

- (iii) $F_n = 2^{2^n} + 1$ defines a set of positive integers called Fermat numbers for $n = 0, 1, 2, \dots$ i.e. $F_0 = 2^{2^0} + 1 = 2^1 + 1 = 3$

5

Use mathematical induction to show that

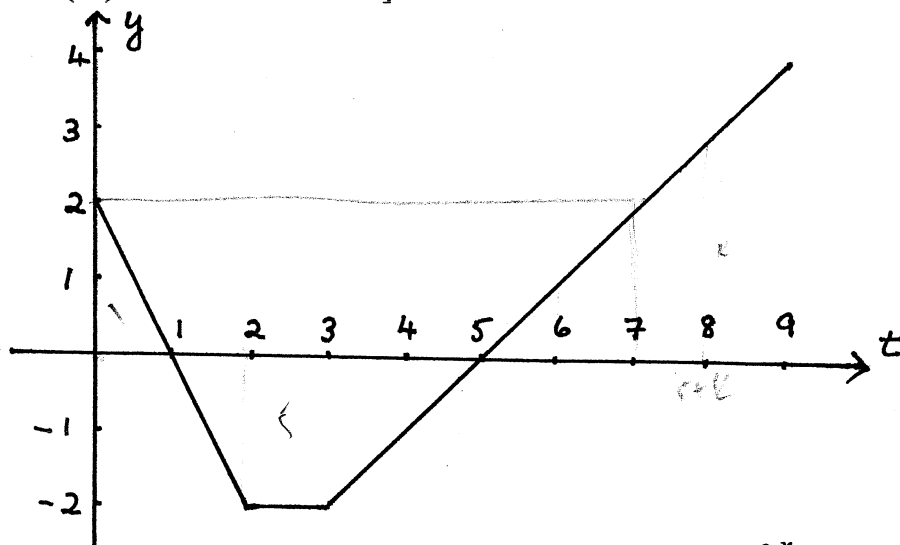
$$F_n = F_0 F_1 F_2 \dots F_{n-1} + 2 \text{ for } n \geq 1$$

- (iv) Hence, or otherwise, show that the highest common factor of any two Fermat numbers is 1.

3

[Hint: Let k be a common factor of F_m and F_n , where $m < n$, and use (i) and (iii) to show that $k = 1$]

b)



5

The graph of $y = f(t)$ $0 \leq t \leq 9$ is shown. If $F(x) = \int_0^x f(t) dt$ find:

- (i) the values of x for which $F(x) = 0$
- (ii) the coordinates of any stationary points.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$(a) \quad y = xe^{-x}$$

$$\frac{dy}{dx} = 1 \cdot e^{-x} + x \cdot (-e^{-x})$$

$$= e^{-x}(1-x)$$

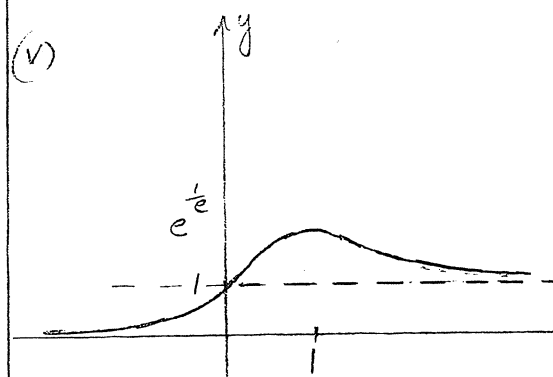
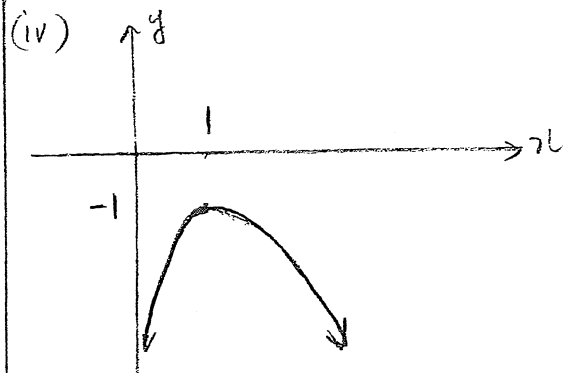
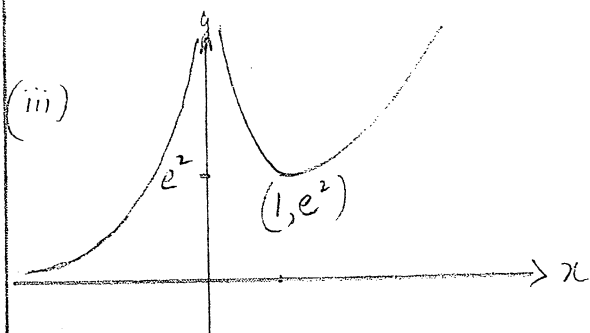
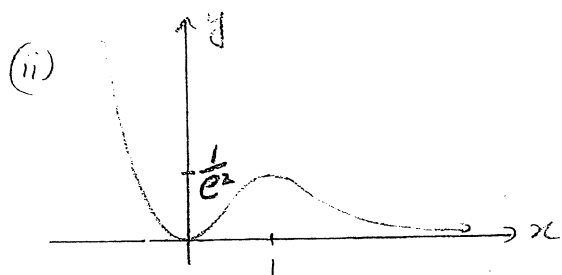
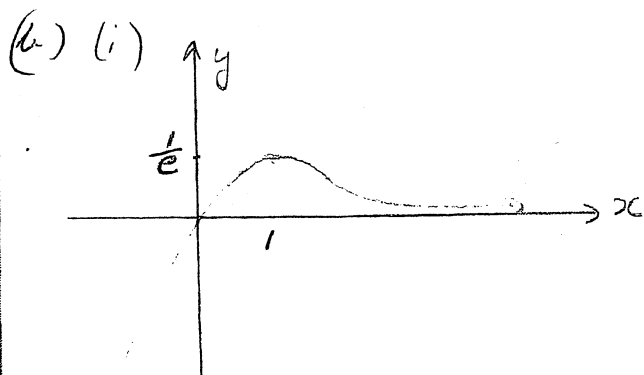
Stat pts when $\frac{dy}{dx} = 0$

$$e^{-x}(1-x) = 0$$

$$x = 1$$

$$y = 1 \cdot e^{-1} = \frac{1}{e}$$

$\therefore (1, \frac{1}{e})$ is a stat pt



$$(c) \quad \frac{3-x}{\sqrt{x-1}} \geq \frac{\sqrt{x-1}}{3-x} \quad \begin{matrix} x > 1 \\ x \neq 3 \end{matrix}$$

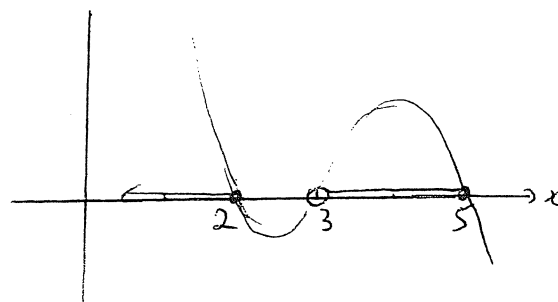
$$\frac{3-x}{\sqrt{x-1}} \geq \frac{(\sqrt{x-1})^2}{3-x}$$

$$\frac{3-x}{1} - \frac{x-1}{3-x} \geq 0$$

$$\frac{9-6x+x^2-x+1}{3-x} \geq 0 \quad \frac{(x(3-x))^2}{(3-x)^2}$$

$$(x^2 - 7x + 10)(3-x) \geq 0$$

$$(x-2)(x-5)(3-x) \geq 0$$



$$1 < x \leq 2 \text{ or } 3 < x \leq 5$$

Question 2

$$\begin{aligned}
 (a) (i) & \int_0^1 \frac{2x}{1+2x} dx \\
 &= \int_0^1 \frac{1+2x-1}{1+2x} dx \\
 &= \int_0^1 \left(1 - \frac{1}{1+2x} \right) dx \\
 &= \left[x - \frac{1}{2} \ln(1+2x) \right]_0^1 \\
 &= \left(1 - \frac{1}{2} \ln 3 \right) - \left(0 - \frac{1}{2} \ln 1 \right) \\
 &= 1 - \frac{1}{2} \ln 3
 \end{aligned}$$

$$\begin{aligned}
 (ii) & \int_2^3 \frac{x+1}{\sqrt{x^2+2x+5}} dx \\
 &= \frac{1}{2} \int_2^3 (2x+2)(x^2+2x+5)^{-1/2} dx \\
 &= \frac{1}{2} \left[\frac{(x^2+2x+5)^{1/2}}{1/2} \right]_2^3 \\
 &= \left[\sqrt{x^2+2x+5} \right]_2^3 \\
 &= \sqrt{20} - \sqrt{13}
 \end{aligned}$$

$$\begin{aligned}
 (iii) & \int_2^4 \frac{dx}{x\sqrt{x-1}} \quad \begin{array}{l} u = \sqrt{x-1} \\ x = u^2 + 1 \\ dx = 2u du \\ x=2 \quad u=1 \\ x=4 \quad u=\sqrt{3} \end{array} \\
 &= \int_1^{\sqrt{3}} \frac{2u du}{(u^2+1) \cdot u} \\
 &= 2 \int_1^{\sqrt{3}} \frac{du}{u^2+1} \\
 &= 2 \left[\tan^{-1} u \right]_1^{\sqrt{3}} \\
 &= 2 \left(\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right) \\
 &= 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\
 &= \frac{\pi}{6}
 \end{aligned}$$

$$(b) \quad x^3 - 2x + 4 = 0 \text{ has roots } \alpha, \beta, \gamma.$$

$$\text{Let } P(x) = x^3 - 2x + 4$$

Eqⁿ with roots $\alpha^2, \beta^2, \gamma^2$ is

$$P(\sqrt{x}) = 0$$

$$(\sqrt{x})^3 - 2\sqrt{x} + 4 = 0$$

$$x\sqrt{x} - 2\sqrt{x} + 4 = 0$$

$$(x-2)\sqrt{x} = -4$$

$$(x-2)^2 \cdot x = 16$$

$$x^3 - 4x^2 + 4x = 16$$

$$x^3 - 4x^2 + 4x - 16 = 0$$

$$\begin{aligned}
 (ii) & x^3 - 4x^2 + 4x - 16 \\
 &= x^2(x-4) + 4(x-4) \\
 &= (x-4)(x^2+4) \\
 &= (x-4)(x+2i)(x-2i)
 \end{aligned}$$

Since $x^3 - 4x^2 + 4x - 16 = 0$ has roots $\alpha^2, \beta^2, \gamma^2$ then

$$\begin{aligned}
 & (x - \alpha^2)(x - \beta^2)(x - \gamma^2) \\
 &= (x-4)(x+2i)(x-2i)
 \end{aligned}$$

Thus, without loss of generality,

$$\alpha^2 = 4 \quad \beta^2 = -2i \quad \gamma^2 = 2i$$

ie only α can be real, β and γ are non-real complex numbers

Question 3

(a) $P(z) = z^7 - 1$

(i) $z^7 - 1 = 0$

$z^7 = 1 \Rightarrow |z| = 1$

Let $z = \cos \theta + i \sin \theta$

$(\cos \theta + i \sin \theta)^7 = 1$

$\cos 7\theta + i \sin 7\theta = 1$

$\cos 7\theta = 1 \quad \sin 7\theta = 0$

$7\theta = 2k\pi$

$\theta = \frac{2k\pi}{7}$

$k=0 \quad z_0 = \cos 0 + i \sin 0 = 1$

$k=1 \quad z_1 = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$

$k=2 \quad z_2 = \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7}$

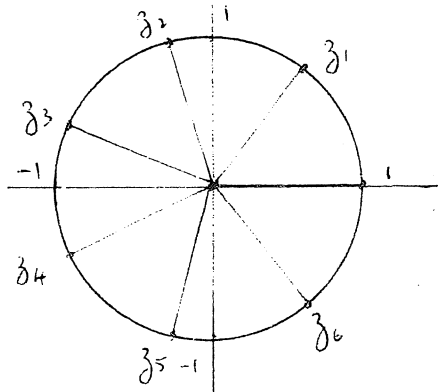
$k=3 \quad z_3 = \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7}$

$k=4 \quad z_4 = \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$
 $= \cos \frac{6\pi}{7} - i \sin \frac{6\pi}{7}$

$k=5 \quad z_5 = \cos \frac{10\pi}{7} + i \sin \frac{10\pi}{7}$
 $= \cos \frac{4\pi}{7} - i \sin \frac{4\pi}{7}$

$k=6 \quad z_6 = \cos \frac{12\pi}{7} + i \sin \frac{12\pi}{7}$
 $= \cos \frac{2\pi}{7} - i \sin \frac{2\pi}{7}$

(ii)



(iii) $P(z) = (z-1)(z-z_1)(z-z_2)(z-z_3)(z-z_4) \times (z-z_5)(z-z_6)$

where $z_1, \dots, z_6$ are indicated in (i)

(iv)

$(z-z_1)(z-z_6) = z^2 - (z_1+z_6)z + z_1z_6$
 $= z^2 - 2\cos \frac{2\pi}{7} z + 1$

Similarly we find $(z-z_2)(z-z_5)$ and $(z-z_3)(z-z_4)$

$\therefore P(z) =$

$(z-1)(z^2 - 2\cos \frac{2\pi}{7} z + 1)(z^2 - 2\cos \frac{4\pi}{7} z + 1) \times (z^2 - 2\cos \frac{6\pi}{7} z + 1)$

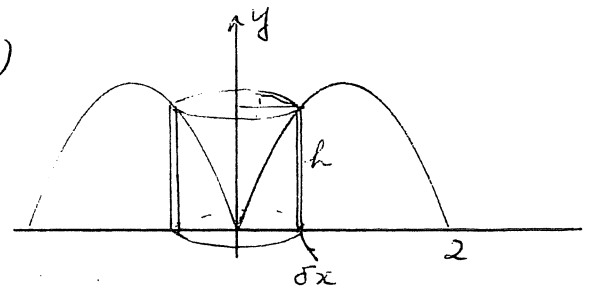
(iv) $z_0 + z_1 + \dots + z_6 = \frac{-\text{coeff } z^6}{\text{coeff } z^7} = 0$

$\therefore 1 + 2\cos \frac{2\pi}{7} + 2\cos \frac{4\pi}{7} + 2\cos \frac{6\pi}{7} = 0$

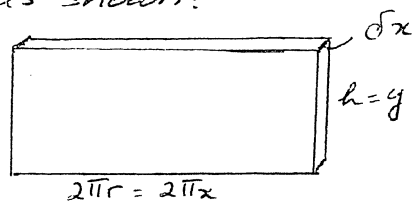
$2(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}) = -1$

$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$

(b)



When the cylindrical shell of radius $r = x$, height $h = y$ and width δx is cut parallel to y axis and "flattened out" its volume δV is approx that of a rectangular prism as shown:



$\delta V = 2\pi x y \delta x$

$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 2\pi x y \delta x = \int_0^2 2\pi x y dx$

(ii) $xy = x \cdot x(2-x) = 2x^2 - x^3$

$V = 2\pi \int_0^2 2x^2 - x^3 dx$
 $= 2\pi \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2$

$= 2\pi \left(\frac{16}{3} - \frac{16}{4} - 0 \right)$

Volume = $\frac{8\pi}{3}$ units³

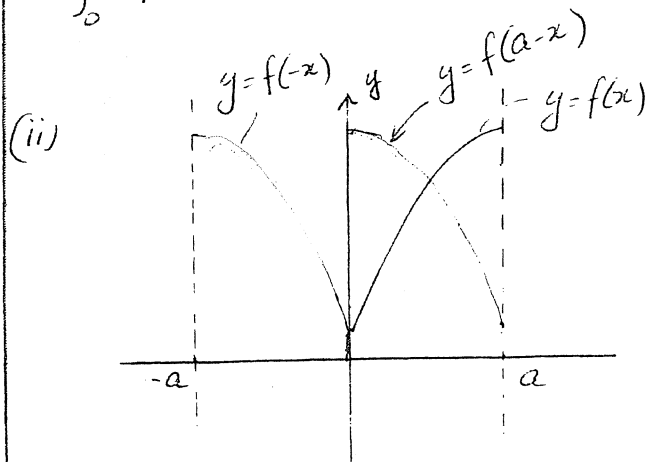
Question 4

(a) (i) $\int_0^a f(a-x) dx$ Let $u = a-x$
 $du = -dx$
 $dx = -du$
 $x=0 \quad u=a$
 $x=a \quad u=0$

$$= \int_a^0 f(u) \cdot (-1) du$$

$$= \int_0^a f(u) du$$

$$= \int_0^a f(x) dx$$



If $y = f(x)$ $0 \leq x \leq a$ is reflected in the y axis we get

$$y = f(-x) \quad -a \leq x \leq 0$$

If this is then translated 'a' units to the right we get

$$y = f(-(x-a)) \quad 0 \leq x \leq a$$

$$= f(a-x) \quad 0 \leq x \leq a$$

From the graph it can be seen that the area under $y = f(x)$ from $x=0$ to $x=a$ ($\int_0^a f(x) dx$) equals the area under $y = f(-x)$ from $x=-a$ to $x=0$ and this is in turn equal to the area under $y = f(a-x)$ from $x=0$ to $x=a$ (ie $\int_0^a f(a-x) dx$)

(iii) $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

$$= \int_0^{\pi/2} \frac{\sqrt{\sin(\frac{\pi}{2}-x)}}{\sqrt{\sin(\frac{\pi}{2}-x)} + \sqrt{\cos(\frac{\pi}{2}-x)}} dx \quad (\text{by (i)})$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

Now

$$\int_0^{\pi/2} \left(\frac{\sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}} + \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} \right) dx$$

$$= \int_0^{\pi/2} 1 dx$$

$$= \left[x \right]_0^{\pi/2} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Since $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$

$$= \frac{\pi}{4}$$

(b) $I_n = \int_0^1 x^n e^{-x} dx \quad n \geq 0$

(i) $= \int_0^1 \frac{d(e^{-x})}{dx} \cdot x^n dx$

$$= \left[-e^{-x} \cdot x^n \right]_0^1 - \int_0^1 -e^{-x} \cdot n x^{n-1} dx$$

$$= (-e^{-1} \cdot 1 - 0) + n \int_0^1 x^{n-1} e^{-x} dx$$

$$= n I_{n-1} - \frac{1}{e}$$

(ii) $\int_0^1 x^3 e^{-x} dx = I_3$

$$= 3 I_2 - \frac{1}{e}$$

$$= 3 \left(2 I_1 - \frac{1}{e} \right) - \frac{1}{e}$$

$$= 6 \left(I_0 - \frac{1}{e} \right) - \frac{4}{e}$$

$$= 6 I_0 - \frac{10}{e}$$

$$I_0 = \int_0^1 e^{-x} dx$$

$$= \left[-e^{-x} \right]_0^1$$

$$= -e^{-1} - (-e^0)$$

$$= 1 - \frac{1}{e}$$

$$\int_0^1 x^3 e^{-x} dx = 6 \left(1 - \frac{1}{e} \right) - \frac{10}{e}$$

$$= 6 - \frac{16}{e}$$

Question 5

$$(c) \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \theta \tan \frac{\theta}{2} d\theta$$

$$t = \tan \frac{\theta}{2} \quad \theta = 2 \tan^{-1} t$$

$$d\theta = \frac{2}{1+t^2} dt$$

$$\theta = \frac{\pi}{3} \quad t = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{2} \quad t = \tan \frac{\pi}{4} = 1$$

$$\cot \theta = \frac{1-t^2}{2t}$$

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \theta \cdot \tan \frac{\theta}{2} d\theta$$

$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{1-t^2}{2t} \cdot t \cdot \frac{2}{1+t^2} dt$$

$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{1-t^2}{1+t^2} dt$$

$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2-(1+t^2)}{1+t^2} dt$$

$$= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2}{1+t^2} - 1 dt$$

$$= \left[2 \tan^{-1} t - t \right]_{\frac{1}{\sqrt{3}}}^1$$

$$= (2 \tan^{-1} 1 - 1) - (2 \tan^{-1} \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}})$$

$$= \left(2 \times \frac{\pi}{4} - 1 \right) - \left(2 \times \frac{\pi}{6} - \frac{1}{\sqrt{3}} \right)$$

$$= \frac{\pi}{6} - 1 + \frac{1}{\sqrt{3}}$$

$$(a) (x+iy)^2 = -12+16i \quad (x, y \text{ real})$$

$$x^2 - y^2 + 2ixy = -12 + 16i$$

$$x^2 - y^2 = -12 \quad \text{--- (1)}$$

$$2xy = 16$$

$$y = \frac{8}{x} \quad \text{--- (2)}$$

Subst (2) in (1)

$$x^2 - \frac{64}{x^2} = -12$$

$$x^4 + 12x^2 - 64 = 0$$

$$(x^2 + 16)(x^2 - 4) = 0$$

$$x = 2, -2 \quad (x \text{ real})$$

$$y = 4, -4$$

$$x = 2, y = 4 \quad \text{or} \quad x = -2, y = -4$$

$$(b) z^3 + (6+i)z^2 + (17+8i)z + 33i + 30 = 0$$

(i) $\alpha = pi$ is a root

$$(pi)^3 + (6+i)(pi)^2 + (17+8i)pi + 33i + 30 = 0$$

$$-p^3i - 6p^2 - ip^2 + 17pi - 8p + 33i + 30 = 0$$

Equating real and imag. parts

$$-p^3 - p^2 + 17p + 33 = 0 \quad \text{--- (1)}$$

$$-6p^2 - 8p + 30 = 0 \quad \text{--- (2)}$$

$$\text{From (2)} \quad 3p^2 + 4p - 15 = 0$$

$$(3p - 5)(p + 3) = 0$$

$$p = \frac{5}{3}, -3$$

Test in (1)

$$p \neq \frac{5}{3} \text{ since } 5 \nmid 33, 3 \nmid 1$$

$$p = -3 \quad \text{LHS} = -(-3)^3 - (-3)^2 + 17 \times -3 + 33$$

$$= 27 - 9 - 51 + 33$$

$$= 0 = \text{RHS}$$

$$\therefore \alpha = -3i$$

$$(ii) \alpha + \beta + \gamma = -(6+i)$$

$$-3i + \beta + \gamma = -6 - i$$

$$\beta + \gamma = -6 + 2i \quad \text{--- (1)}$$

$$\alpha\beta\gamma = -(33i + 30)$$

$$-3i\beta\gamma = -3i(11 - 10i)$$

$$\beta\gamma = 11 - 10i \quad \text{--- (2)}$$

From (1) $x = -6 + 2i - \beta$

Subst in (2)

$$\beta(-6 + 2i - \beta) = 11 - 10i$$

$$(-6 + 2i)\beta - \beta^2 = 11 - 10i$$

$$\beta^2 + (6 - 2i)\beta + 11 - 10i = 0$$

$$\beta^2 + (6 - 2i)\beta + (3 - i)^2 = -11 + 10i + (3 - i)^2$$

$$(\beta + (3 - i))^2 = -11 + 10i + 9 - 6i - 1$$

$$= -3 + 4i$$

$$\beta + (3 - i) = \pm \sqrt{-3 + 4i} \quad (*)$$

$$= \pm (1 + 2i)$$

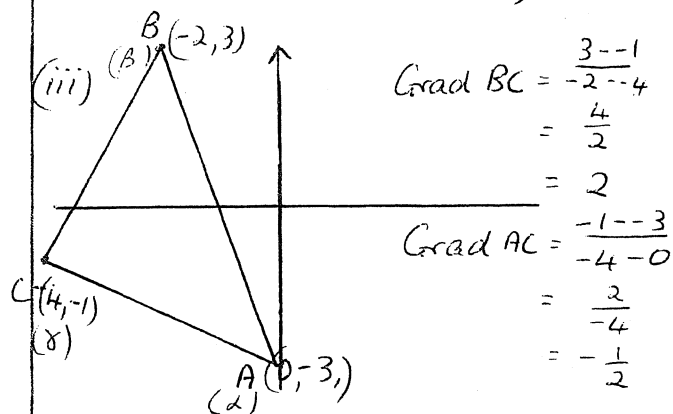
$$\beta = -3 + i + 1 + 2i, -3 + i - 1 - 2i$$

$$= -2 + 3i, -4 - i$$

$$* \sqrt{-12 + 16i} = 2\sqrt{-3 + 4i}$$

$$\pm (2 + 4i) = 2\sqrt{-3 + 4i}$$

$$\therefore \sqrt{-3 + 4i} = \pm (1 + 2i)$$



$$\text{Grad BC} \times \text{Grad AC} = -\frac{2}{3} \times \frac{3}{2} = -1$$

$\therefore A, B, C$ vertices of right-angled Δ

$$(c) (z - \bar{z})^2 + 8(z + \bar{z}) = 16$$

Let $z = x + iy$

$$\therefore \bar{z} = x - iy$$

$$z - \bar{z} = 2iy$$

$$z + \bar{z} = 2x$$

$\therefore$ Eqⁿ becomes

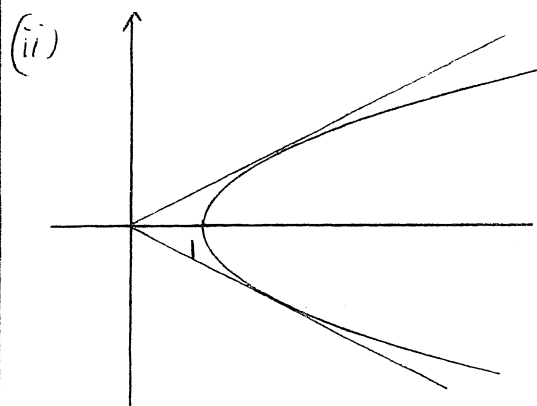
$$(2iy)^2 + 8 \cdot 2x = 16$$

$$-4y^2 + 16x = 16$$

$$16x = 16 + 4y^2$$

$$x = 1 + \frac{y^2}{4}$$

which is a parabola.



Maximum and minimum values of $\arg z$ occur when $y = mx$ is a tangent to $x = 1 + \frac{y^2}{4}$

$$y = mx \quad \text{--- (1)}$$

$$x = 1 + \frac{y^2}{4} \quad \text{--- (2)}$$

Subst (1) in (2)

$$x = 1 + \frac{(mx)^2}{4}$$

$$4x = 4 + m^2 x^2$$

$$m^2 x^2 - 4x + 4 = 0$$

Line is a tangent when quadratic has equal roots i.e. $\Delta = 0$

$$(-4)^2 - 4 \times m^2 \times 4 = 0$$

$$16 - 16m^2 = 0$$

$$m^2 = 1$$

$$m = \pm 1$$

$$\tan \theta = \pm 1$$

$$\theta = \pm \frac{\pi}{4}$$

$$\therefore -\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$$

Question 6

$$(a) \frac{x^2}{9} + \frac{y^2}{5} = 1$$

$$a^2 = 9 \quad b^2 = 5$$

$$a = 3 \quad b = \sqrt{5}$$

$$(i) b^2 = a^2(1 - e^2)$$

$$5 = 9(1 - e^2)$$

$$e^2 = \frac{4}{9}$$

$$e = \frac{2}{3} \quad (e > 0)$$

$$(ii) ae = 3 \times \frac{2}{3} = 2$$

$$S(2, 0) \quad S'(-2, 0)$$

$$(iii) \frac{a}{e} = \frac{3}{\frac{2}{3}} = \frac{9}{2}$$

$$\text{Directrices: } x = \frac{9}{2}, \quad x = -\frac{9}{2}$$

$$(iv) \frac{x^2}{9} + \frac{y^2}{5} = 1$$

$$\frac{2x}{9} + \frac{2y}{5} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{9} \times \frac{5}{2y}$$

$$= -\frac{5x}{9y}$$

$$\text{At } P(x_1, y_1) \quad \frac{dy}{dx} = -\frac{5x_1}{9y_1}$$

Eqⁿ of tangent at P is

$$y - y_1 = -\frac{5x_1}{9y_1}(x - x_1)$$

$$\frac{yy_1}{5} - \frac{y_1^2}{5} = -\frac{xx_1}{9} + \frac{x_1^2}{9}$$

$$\frac{xx_1}{9} + \frac{yy_1}{5} = \frac{x_1^2}{9} + \frac{y_1^2}{5}$$

$$= 1 \quad (\text{since } P(x_1, y_1) \text{ lies on ellipse})$$

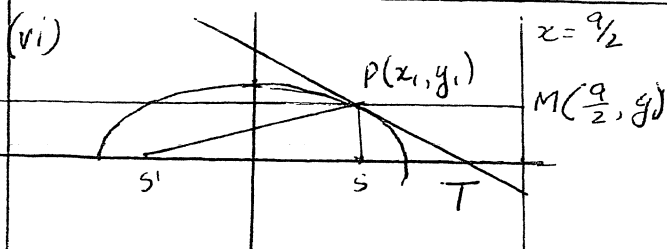
$$\frac{xx_1}{9} + \frac{yy_1}{5} = 1$$

$$(v) \text{ When } y = 0$$

$$\frac{xx_1}{9} = 1$$

$$x = \frac{9}{x_1}$$

$$\therefore T \text{ has coords } \left(\frac{9}{x_1}, 0\right)$$

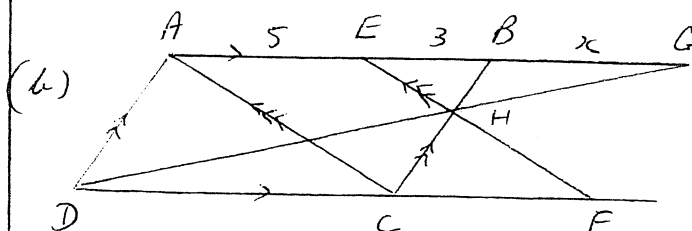


Let $M(\frac{9}{2}, y_1)$ and $M'(-\frac{9}{2}, y_1)$ be the feet of the perpendiculars from P to the directrices

$$\frac{PS}{PM} = \frac{PS'}{PM'} (= e) \quad (\text{focus-directrix definition})$$

$$\therefore \frac{PS}{PS'} = \frac{PM}{PM'} = \frac{\frac{9}{2} - x_1}{\frac{9}{2} + x_1} = \frac{9 - 2x_1}{9 + 2x_1}$$

$$\begin{aligned} \text{Now } \frac{TS}{TS_1} &= \frac{\frac{9}{x_1} - 2}{\frac{9}{x_1} + 2} \\ &= \frac{9 - 2x_1}{9 + 2x_1} \\ &= \frac{PS}{PS'} \end{aligned}$$



$$\text{Let } GB = x$$

$$\text{In } \triangle BAC$$

$$\frac{BH}{HC} = \frac{BE}{EA} \quad (\text{line parallel to one side of } \triangle \text{ divides other 2 sides in same ratio})$$

$$\therefore \frac{BH}{BC} = \frac{3}{8}$$

$$\text{In } \triangle GAD, \triangle GBH$$

$$\angle BGH \text{ is common}$$

$$\angle GAD = \angle GBH \quad (\text{corresp } \angle \text{s equal } BH \parallel AD)$$

$$\triangle GAD \parallel \triangle GBH \quad (\text{equiangular})$$

$$\therefore \frac{GA}{GB} = \frac{AD}{BH} = \frac{BC}{BH} \quad (BC = AD \text{ opp sides of parallelogram})$$

$$\frac{x+8}{x} = \frac{8}{3}$$

$$3x + 24 = 8x$$

$$24 = 5x$$

$$x = 4.8$$

$$\therefore AG = 12.2 \text{ cm}$$

$$(ii) \text{ Area } \triangle CDH = \frac{1}{2} \times CH \times DC \sin \hat{DCH}$$

$$\text{Area } ABCD = BC \times CD \sin \hat{DCH}$$

$$\frac{\text{Area } \triangle CDH}{\text{Area } ABCD} = \frac{\frac{1}{2} \cdot CH \cdot DC \cdot \sin \hat{DCH}}{BC \times DC \cdot \sin \hat{DCH}}$$

$$= \frac{\frac{1}{2} CH}{BC}$$

$$= \frac{1}{2} \times \frac{5}{8}$$

$$= \frac{5}{16}$$

Question 7

(a) (i) $\frac{d(\frac{1}{2}v^2)}{dx} = 2x^3 - 10x$

$$\frac{1}{2}v^2 = \frac{2x^4}{4} - \frac{10x^2}{2} + C$$

When $x=0$ $v=u$

$$\frac{1}{2}u^2 = C$$

$$\frac{1}{2}v^2 = \frac{x^4}{2} - \frac{10x^2}{2} + \frac{u^2}{2}$$

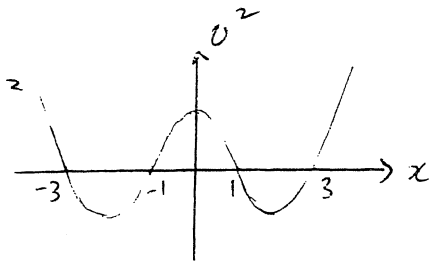
$$v^2 - u^2 = x^4 - 10x^2$$

(ii) (2) If $u=3$ then $v^2 - 9 = x^4 - 10x^2$

$$v^2 = x^4 - 10x^2 + 9$$

$$= (x^2 - 1)(x^2 - 9)$$

$$= (x-1)(x+1)(x-3)(x+3)$$



Since $v^2 \geq 0$ then $(x-1)(x+1)(x-3)(x+3) \geq 0$

$$\therefore x \leq -3 \text{ or } -1 \leq x \leq 1 \text{ or } x \geq 3$$

Since particle starts at $x=0$ with $v=3$ it then moves to the right (positive direction). At $x=1$ $v=0$ and $\ddot{x} = -8$ and so particle will then move to the left until it reaches $x=-1$ where $v=0$ and $\ddot{x} = 8$. This means particle will then move to the right until $v=0$ again at $x=1$. Thus particle oscillates between $x=1$ and $x=-1$ ie within the interval $-1 \leq x \leq 1$

(β) Not SHM since $\ddot{x} \neq -\omega^2(x-a)$

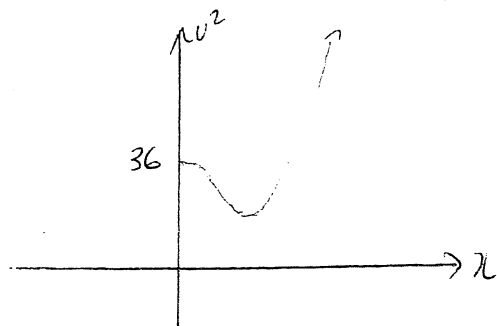
(iii) If $u=6$ then $v^2 - 36 = x^4 - 10x^2$

$$v^2 = x^4 - 10x^2 + 25 + 11$$

$$= (x^2 - 5)^2 + 11$$

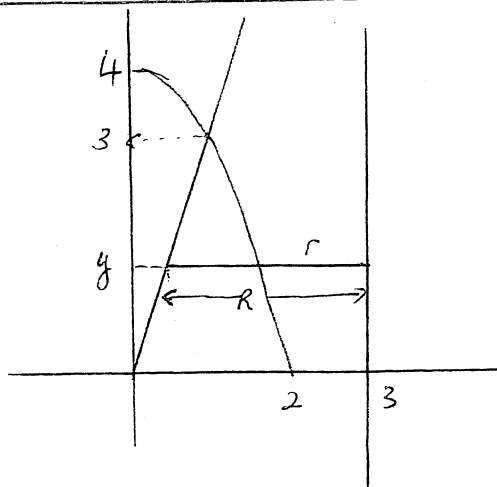
$$\therefore v^2 > 0 \text{ for all } x \text{ in domain}$$

$\therefore$ Particle will never stop



Moves to the right with decreasing velocity until it passes $x=\sqrt{5}$ (with velocity $\sqrt{11}$) and continues to the right with increasing velocity

(4)



$$\begin{aligned}
 y &= 4 - x^2 \\
 y &= 3x \\
 x^2 + 3x - 4 &= 0 \\
 (x+4)(x-1) &= 0 \\
 x &= -4, 1 \\
 y &= -12, 3
 \end{aligned}$$

Let $A(y)$ be the area of the cross-section at height y

$$A(y) = \pi R^2 - \pi r^2$$

$$R = 3 - \frac{y}{3}$$

$$y = 3x$$

$$r = 3 - \sqrt{4-y}$$

$$y = 4 - x^2$$

$$R^2 = \left(3 - \frac{y}{3}\right)^2 = 9 - 2y + \frac{y^2}{9}$$

$$r^2 = \left(3 - \sqrt{4-y}\right)^2 = 9 - 6\sqrt{4-y} + 4 - y$$

$$\begin{aligned}
 A(y) &= \pi \left[\left(9 - 2y + \frac{y^2}{9}\right) - (9 - 6\sqrt{4-y} + 4 - y) \right] \\
 &= \pi \left(\frac{y^2}{9} + y + 6\sqrt{4-y} - 4 \right)
 \end{aligned}$$

$$V = \int_0^3 \pi \left(\frac{y^2}{9} + y + 6\sqrt{4-y} - 4 \right) dy$$

$$= \pi \left[\frac{y^3}{27} + \frac{y^2}{2} + \frac{6(4-y)^{3/2}}{\frac{3}{2} \times -1} - 4y \right]_0^3$$

$$= \pi \left\{ \left(\frac{27}{27} + \frac{9}{2} - 4 \times 1 - 12 \right) - \left(-4 \times 4^{3/2} \right) \right\}$$

$$= \pi \left\{ \frac{41}{2} - 16 + 32 \right\}$$

$$= \frac{43\pi}{2} \quad 12\frac{1}{2}\pi.$$

OR cylindrical shells.

Question 8

(a)(i) If x is a factor of both y & z then $y = ax$ and $z = bx$ where a, b are integers

$$z - y = bx - ax = (b - a)x$$

ie x is a factor of $z - y$

(ii) $2^{2^{k+1}} = 2^{2^k \times 2} = (2^{2^k})^2$

(iii) $F_n = 2^{2^n} + 1 \quad n = 0, 1, 2, \dots$
 Let $F_n = F_0 F_1 \dots F_{n-1} + 2 \quad n \geq 1$

$$\begin{aligned} F_0 &= 3 \\ n=1 \quad F_1 &= 2^{2^1} + 1 \\ &= 2^2 + 1 \\ &= 5 \\ &= 3 + 2 \\ &= F_0 + 2 \end{aligned}$$

$\therefore$ Proposition is true for $n=1$

Let k be a value for which proposition is true ($k \geq 1$)

ie $F_k = F_0 F_1 \dots F_{k-1} + 2$

Aim to show that proposition is then true for $n = k+1$

ie $F_{k+1} = F_0 F_1 \dots F_k + 2$

$$\begin{aligned} F_{k+1} &= 2^{2^{k+1}} + 1 \\ &= (2^{2^k})^2 + 1 \\ &= (F_k - 1)^2 + 1 \\ &= F_k^2 - 2F_k + 1 + 1 \\ &= F_k(F_k - 2) + 2 \\ &= F_k(F_0 F_1 \dots F_{k-1}) + 2 \quad (\text{by inductive hypothesis}) \\ &= F_0 F_1 \dots F_{k-1} F_k + 2 \end{aligned}$$

which is of the required form
 $\therefore$ If proposition is true for $n=k$ it is also true for $n=k+1$. Since it

is true for $n=1$ it is also true for $n=1+1=2$ and hence by induction it is true for all positive integers

(iv) If $m < n$ then

$$F_n = F_0 F_1 \dots F_m F_{n-1} + 2$$

Let k be the hcf of F_n and F_m

$\therefore k$ is a factor of F_n and $F_0 F_1 \dots F_m F_{n-1}$

Hence k is a factor of their difference

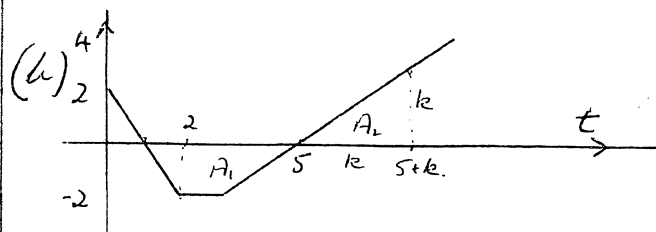
ie k is a factor of 2

ie $k = 1$ or 2

But $F_n = 2^{2^n} + 1$ is odd

$$\therefore k \neq 2$$

$$\text{ie } k = 1$$



(i) $F(x) = 0$ when $x = 0, 2, 5+k$ where k is such that $A_1 = A_2$

$$A_1 = \frac{1}{2} (1+3) \times 2 = 4$$

$$A_2 = \frac{1}{2} k^2$$

$$\therefore k^2 = 8$$

$$k = \sqrt{8} = 2\sqrt{2}$$

ie $x = 0, 2, 5 + 2\sqrt{2}$

(ii) Stationary points at

$$x = 1 \quad \text{and} \quad x = 5$$

$$F(1) = \frac{1}{2} \times 1 \times 2 = 1 \quad F(5) = -4$$

$\therefore (1, 1)$ and $(5, -4)$ are stationary points